

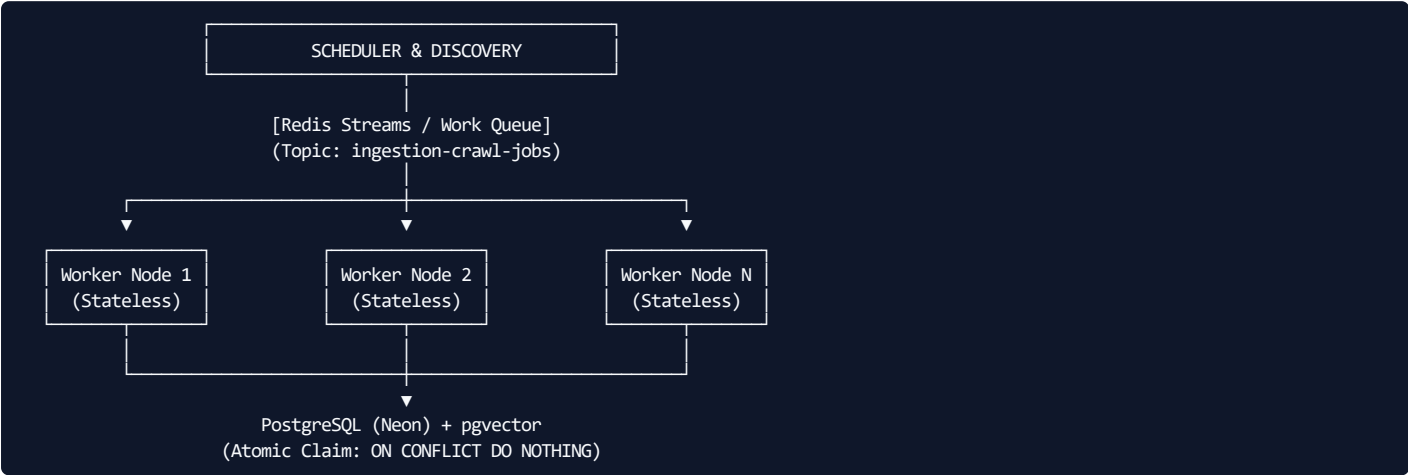
GraphOne AI Intelligence Search — System Architecture Specification

Document Version: 1.0 | Target Submission Artifact: architecture.pdf (≤3 Pages) | Classification: Production Engineering Blueprint

System Scope: High-Throughput Structured Multi-Entity Ingestion, Entity Resolution, Verification & Freshness Intelligence

1. Scale Strategy (500,000+ Records Architectural Scaling)

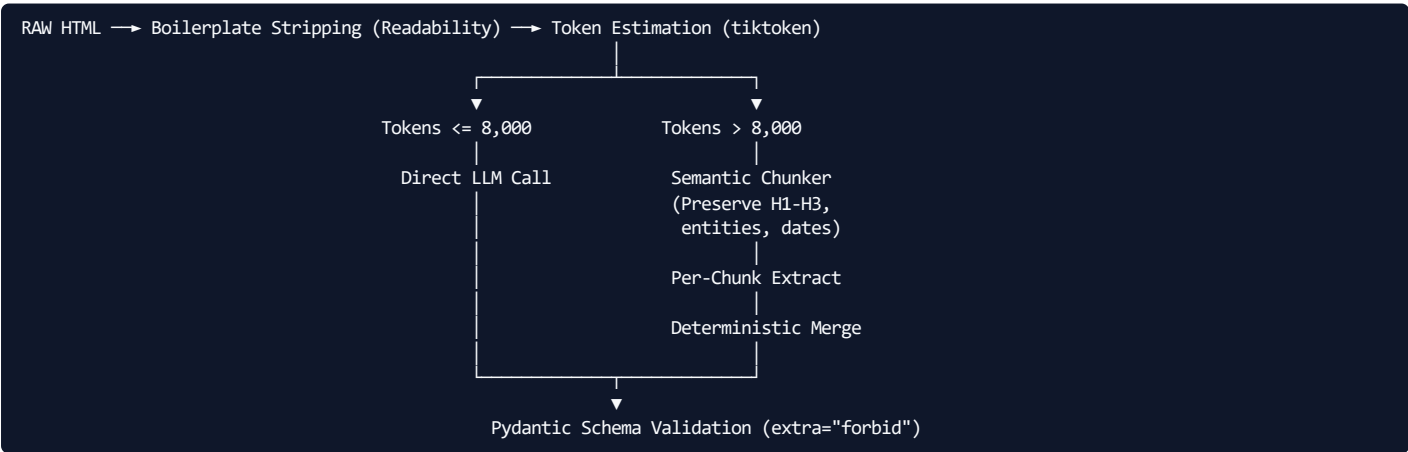
GraphOne scales from seed ingestion to 500,000+ records without code refactoring through horizontal worker partitioning and stateless extraction:



- **Stateless Workers & Queue Coordination:** Crawlers run in autonomous worker pools driven by Redis Streams consumer groups.
- **Atomic Claim & Database Idempotency:** Workers claim URLs atomically via PostgreSQL `ON CONFLICT DO NOTHING`. Uniqueness constraints on `canonical_url`, `arxiv_id`, and normalized entities prevent duplicate work across distributed workers.
- **Object Storage Offloading:** Raw HTML/XML payloads are stored in Cloudflare R2 object storage with sha256 content hashes, storing only URI pointers and metadata in PostgreSQL to prevent database bloat.

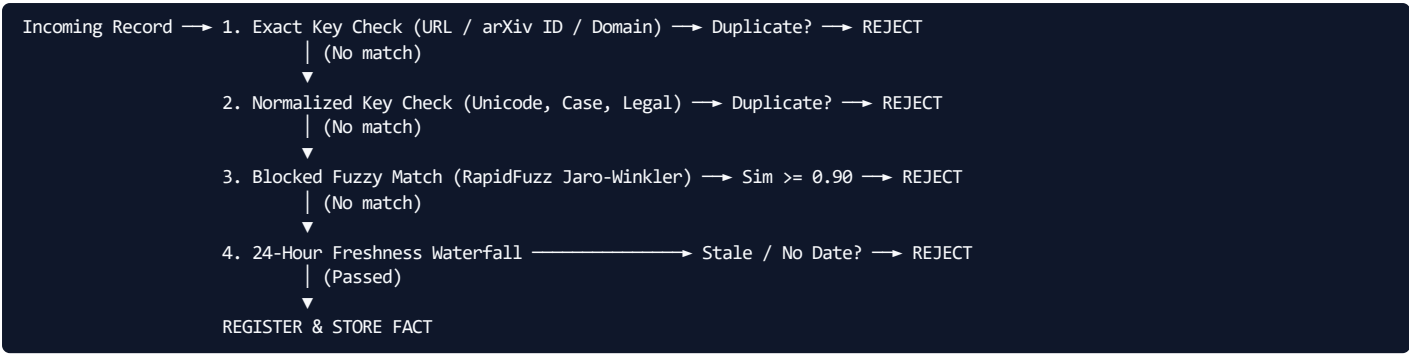
2. 413 & 429 Resiliency Engine

Deterministic handling of LLM context limits (413) and rate limits (429):



- **Payload Too Large (413) Semantic Chunking:** Content exceeding 8,192 tokens is stripped of boilerplate and segmented along paragraph/heading boundaries without splitting entities or numbers. Chunks are extracted individually and merged deterministically.
- **Rate Limit (429) & Multi-Tier Fallback Chain:** Provider fallback priority: `Gemini 2.0 Flash` → `Groq Llama 3.3 70B` → `DeepSeek-V3`. Uses exponential backoff with full jitter and honors HTTP `Retry-After` headers.
- **Dead-Letter Queue (DLQ):** Requests failing 3 attempts are routed to `data_quality_flags` with `status: PENDING_REVIEW`.

3. Freshness & Deduplication Across Distributed Nodes



- **Three-Tier Deduplication Pipeline:** 1. *Tier 1 (Exact Key):* $O(1)$ lookup on canonical URL, arXiv ID, or official domain. 2. *Tier 2 (Normalized Key):* NFKD unicode normalization, lowercasing, punctuation stripping, and legal suffix removal (Inc., Ltd., PBC, LLC). 3. *Tier 3 (Blocked Fuzzy Match):* Blocks records by prefix and entity type (STARTUP:op, JOB:en) and runs RapidFuzz Jaro-Winkler (≥ 0.90) strictly within blocks to prevent $O(n^2)$ degradation.
- **24-Hour Freshness Waterfall:** Date extraction follows: $\text{JSON-LD} \rightarrow \text{OpenGraph/Meta} \rightarrow \text{time datetime} \rightarrow \text{Relative Text} \rightarrow \text{RSS Timestamp}$. If no timestamp within 24h is verified, the record is rejected.

4. Storage & Database Architecture Justification

POSTGRESQL 16 (NEON SERVERLESS)		
RELATIONAL TABLES (Startups, Jobs, Papers, Lineage, Data Quality)	PGVECTOR EXTENSION (1536-dim embeddings, HNSW cosine index)	GRAPH (Relationships & Aliases)

- **Unified PostgreSQL with pgvector:** Eliminates dual-write lag between separate vector/graph DBs. Provides ACID atomicity across relational metadata, vector embeddings, and entity relationships in a single database engine.
- **Hybrid Lexical + Semantic Search:** Combines full-text search (tsvector + GIN) with vector search (pgvector cosine distance) via Reciprocal Rank Fusion (RRF).
- **Auditable Data Lineage:** Every fact is linked via foreign keys to immutable raw_documents and lineage_events.

5. Data Quality Scoring & Corroboration Framework

Every stored record is assigned a composite Data Quality Score (0–100):

$$DQS = 0.30 \cdot S_{\text{reliability}} + 0.25 \cdot S_{\text{corroboration}} + 0.25 \cdot S_{\text{freshness}} + 0.20 \cdot S_{\text{confidence}} - P_{\text{dispute}}$$

Scoring Component	Weight	Scoring Tiers & Criteria
Source Reliability	30%	100 (Official APIs/arXiv/Greenhouse), 92 (Primary RSS/TechCrunch), 85 (Curated GitHub Directories), 70 (General Web)
Corroboration	25%	100 (Cross-verified across ≥2 independent sources), 80 (Authoritative primary registry single-source), 70 (Curated directory single-source), 50 (Unverified / missing fields)
Freshness	25%	100 (≤24 hours), 85 (≤7 days), 70 (≤30 days), 50 (>30 days / unknown)
Extraction Confidence	20%	100 (Structured API/Exact Regex), 90 (JSON-LD / Meta Tags), 80 (HTML CSS Selector), 70 (Fuzzy / LLM)
Dispute Penalty	Deduction	-25 points penalty if independent sources report conflicting values (status=DISPUTED, both values retained with citations)